Trade Shocks and Political Change in Local Labor Markets

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1 Summary

Autor et al. (2013) (hereafter ADH), "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," examine how rising Chinese import competition affected U.S. local labor markets between 1990 and 2007. Their primary research question is whether greater exposure to Chinese imports contributed to declines in manufacturing employment, wages, and labor force participation across U.S. regions. Using commuting zones (CZs) as local labor market units, ADH use a shift-share research design that relates changes in local outcomes to differences in baseline industry composition and subsequent industry-level import growth.

Our project extends this framework to political outcomes. We ask whether CZs more exposed to the China shock experienced different changes in Republican presidential vote margins, and whether these effects appear in the short run, medium run, or long run. To answer this question, we combine the ADH replication package with county-level presidential election returns and historical county-to-CZ crosswalks. Our extension finds suggestive evidence that trade-exposed places experienced later rightward shifts in presidential voting. However, the interpretation is cautious: shift-share designs rely on nontrivial assumptions about initial industry shares and import-growth shocks, and CZ-level political change may reflect belief updating, turnout and registration, elite political supply, or population composition rather than a single mechanism.

2 Introduction

Our project's research question is does exposure to the China trade shock affect political outcomes in the United States? If so, how do these effects evolve over time? The project examines whether commuting zones that were more exposed to rising Chinese import competition experienced different changes in Republican presidential vote margins than less-exposed commuting zones. The question is not only whether trade exposure mattered politically, but also when its effects appeared. For example, we distinguish short-run effects around the Great Recession, medium-run effects around the 2016 election, and longer-run effects that may reflect later shocks such as COVID-19 and inflation.

Understanding this question is important because globalization may reshape political behavior along with the economy. If local labor-market losses from import competition contributed to changes in voting patterns, then trade policy may have broader consequences for democratic representation, polarization, and public support for protectionism. This may help explain why places hit hardest by manufacturing decline may respond differently to national political messages about trade, immigration, redistribution, and economic nationalism.

The question is also important because it connects local economic shocks to national political outcomes. A trade shock that lowers manufacturing employment may eventually influence turnout, party support, media consumption, and attitudes toward government intervention in affected regions. If these effects are persistent, then the consequences of globalization may extend beyond the original employment losses. Thus, studying these dynamics can help policymakers understand the extent of the effect of trade policy and adjustment assistance on political life.

A key empirical challenge is that identification relies on both the measurement of exposure and the validity of the shift-share design. Principally, the ADH exposure measure may suffer from aggregation and temporal bias. Import exposure is aggregated across broad industries, which may conceal substantial heterogeneity at more disaggregated levels. If variation in exposure is driven by a small subset of industries, such as textiles, furniture, and electronics, estimated effects may reflect industry-specific shocks rather than a general China-shock effect.

There are concerns regarding the validity of the Bartik instrument and initial industry shares. Identification depends not only on Chinese export growth but also on baseline industry composition in 1990. These industry shares are not randomly assigned across commuting zones and may be correlated with unobserved determinants of political outcomes, including automation exposure, unionization, education, demographic structure, or pre-existing political trends. Thus, both the IV strategy and reduced-form exposure design require careful discussion of what variation identifies the estimates.

Additionally, there is potential compositional bias due to migration across commuting zones. If trade-exposed regions experience selective in- or out-migration, observed political changes may reflect shifts in population composition instead of changes in individual political preferences. Measurement issues can also arise from mapping county-level election data into historical CZ boundaries and from sensitivity to alternative spatial definitions of labor markets. Together, these concerns motivate our robustness checks, measurement diagnostics, and cautious interpretation.

3 Data

Our analysis combines four primary datasets to construct commuting-zone-level political and trade exposure measures. We use the Autor et al. (2013) replication package for baseline commuting-zone trade exposure measures and pre-determined labor market controls, including industry composition and demographic structure.

To align modern political data with historical labor markets, we use Ferrara et al. (2024), *Crosswalks for U.S. Counties and Congressional Districts, 1790–2020*, which maps contemporary counties to historical boundaries and provides county centroid coordinates. Our preferred crosswalk uses M5 weights where available and M2 weights as fallback for source counties where M5 weights are undefined or zero.

We aggregate counties into 1990 commuting zones using the U.S. Department of Agriculture Economic Research Service Commuting Zones and Labor Market Areas dataset, consistent with the Tolbert and Sizer (1996) classification and the USDA ERS commuting-zone data (U.S. Department of Agriculture Economic Research Service 2026).

Political outcomes are drawn from Amlani and Algara (2021), *Partisanship & Nationalization in American Elections, 1872–2020*, which provides county-level presidential election returns used to construct Republican vote margins.

We follow the sample design of Autor, Dorn, and Hanson (2013). The unit of analysis is the 1990 commuting zone (CZ) of Tolbert and Sizer (1996), covering 722 CZs over 1990–2000 and 2000–2007 stacked in first differences (1,444 observations). Table 1 reports population-weighted means from ADH's replication package. Two findings stand out. Chinese imports per worker (10-year equivalent) more than doubled, from \$1.14k in 1990–2000 to \$2.63k in 2000–2007, tracking China's WTO accession in 2001. Manufacturing employment as a share of population declined from 12.69% in 1990 to 8.51% in 2007, while SSDI receipt increased over the same period.

Table 1: Descriptive statistics for selected CZ-level variables.

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Variable	1990	2000	2007
Δ imports from China per worker (\$k, 10-yr equiv.)	1.14 (0.99)	2.63 (2.01)	—
Manufacturing employment / pop. (%)	12.69 (4.80)	10.51 (4.45)	8.51 (3.60)
Non-manufacturing employment / pop. (%)	57.75 (5.91)	59.16 (5.24)	61.87 (4.95)
Unemployment / pop. (%)	4.80 (0.99)	4.28 (0.93)	4.87 (0.90)
Not in labor force / pop. (%)	24.76 (4.34)	26.05 (4.39)	24.75 (3.70)
SSDI recipients / pop. (%)	1.86 (0.63)	2.75 (1.04)	3.57 (1.41)

Notes: N = 722 commuting zones for each year. Means are weighted by start-of-period CZ population. Standard deviations in parentheses. Variables follow Autor, Dorn & Hanson (2013) Appendix Table 2; all values reproduced from the official replication package.

4 Analysis and Findings

4.1 Identification Strategy and its Plausibility

OLS estimation of the labor-market effect of Chinese import exposure faces an endogeneity problem. Observed U.S. imports reflect both Chinese supply-side and U.S. demand shocks, biasing OLS toward zero. Autor et al. (2013) report an OLS estimate of −0.171 versus an IV estimate of −0.596.

They address this with a Bartik-style instrument, replacing ΔIPW_{uit} with ΔIPW_{oit}, built from same-period Chinese exports to eight other high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland). The identifying assumption is that within-industry import growth shared by these countries reflects China's supply-side improvements, not correlated demand:

$$\Delta IPW_{uit} = \sum_j \left(\frac{L_{ijt}}{L_{ujt}}\right) \cdot \left(\frac{\Delta M_{ucjt}}{L_{it}}\right),$$

$$\Delta IPW_{oit} = \sum_j \left(\frac{L_{ij,t-1}}{L_{uj,t-1}}\right) \cdot \left(\frac{\Delta M_{ocjt}}{L_{i,t-1}}\right).$$

The second-stage equation is estimated as stacked first differences:

$$\Delta L_{itm} = \gamma_t + \beta_1 \Delta IPW_{uit} + X_{it}'\beta_2 + e_{it},$$

where X_{it} contains start-of-period manufacturing share, demographic controls, the routine and offshorability indices of Autor and Dorn (2013), and Census division dummies.

Three pieces of evidence support the strategy. First, the first stage is strong: the instrument predicts the endogenous regressor with coefficient 0.631 ($t = 6.90$), well above the weak-IV threshold. Second, a falsification test shows that 1970s–1980s manufacturing changes do not predict future Chinese exposure, ruling out a pre-existing decline. Third, dropping suspect industries leaves the coefficient near -0.596.

The main residual concern is correlated demand shocks across high-income economies. A gravity-based check yields similar estimates; Chinese manufacturing TFP grew about 8% per year over 1998–2007, against 3.9% in the U.S., supporting the supply-driven interpretation.

4.2 Reproduction of Published Results

We verify that the official replication package functions correctly. Our target is Table 3 of Autor et al. (2013), the 2SLS regression of CZ-level change in manufacturing employment share on ΔIPW_{uit}, instrumented by the eight-country measure. We focus on column (6) with full controls.

Running the official Stata code on the published workfile, we obtain coefficients matching the published values exactly to three decimal places across all six columns. In column (6), the coefficient on ΔIPW is -0.596 (SE 0.099); the first-stage coefficient is 0.631 ($t = 6.90$); $N = 1{,}444$ with $R^2 = 0.343$. Table 2 reports the full comparison.

Table 2: Replication of ADH (2013) Table 3: effect of Chinese import exposure on CZ manufacturing employment share.

Table 2. Replication of ADH (2013) Table 3: Effect of Chinese import exposure on CZ manufacturing employment

	(1)	(2)	(3)	(4)	(5)	**(6)**
Panel A. ADH (2013) published estimates						
Δ imports from China / worker	−0.746	−0.610	−0.538	−0.508	−0.562	−0.596
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
Panel B. Our replication						
Δ imports from China / worker	**−0.746**	**−0.610**	**−0.538**	**−0.508**	**−0.562**	**−0.596**
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
First-stage β (instrument)	0.792	0.664	0.652	0.635	0.638	0.631
	(0.080)	(0.089)	(0.094)	(0.094)	(0.091)	(0.091)
Start-of-period mfg. share		Y	Y	Y	Y	Y
Census division dummies			Y	Y	Y	Y
Demographic controls				Y		Y
Routine / offshorability					Y	Y
Observations	1,444	1,444	1,444	1,444	1,444	1,444
R^2	0.066	0.157	0.282	0.321	0.343	0.343

Notes: 2SLS estimates. Dependent variable is the 10-year-equivalent change in CZ manufacturing employment as a share of working-age population (in percentage points). The endogenous regressor Δ imports from China per worker is instrumented by Chinese imports to eight other high-income countries, weighted by lagged industry-employment shares. Stacked first differences for 1990–2000 and 2000–2007 (N = 722 × 2). Models weighted by start-of-period CZ population share. Standard errors in parentheses, clustered at the state level. Replication coefficients match the published values exactly to three decimal places across all six columns.

Economically, -0.596 implies that a \$1,000 per-worker increase in decadal Chinese exposure lowers a CZ's manufacturing share by 0.6 percentage points. ADH attribute 44% of the 1990–2007 manufacturing decline to this channel. Because we use the original data and code, this is a computational reproduction: the exact match confirms our codebase functions correctly. The next section applies the same strategy to CZ-level presidential political outcomes.

4.3 Extension: Dynamic Political Effects of the China Shock

We extend the ADH local-labor-market design to presidential political outcomes. The outcome is the Republican presidential vote margin in CZ c and election year t. The preferred specification includes CZ fixed effects, election-year fixed effects, the ADH 1990–2007 import-exposure measure interacted with election-year indicators, and time-varying effects of baseline controls. The omitted year is 1988, so coefficients describe how the relationship between China-shock exposure and Republican margin changes relative to the pre-shock election immediately before the main exposure period.

A central measurement challenge is that electoral data are observed for counties whose boundaries change over time, while ADH exposure is measured for 1990 commuting zones. We therefore bridge county election returns to 1990 counties using the FTZ county crosswalks, aggregate counties to 1990 CZs using the Tolbert-Sizer/USDA crosswalk, and merge those outcomes to ADH exposure. Our preferred bridge uses FTZ M5 weights where available and M2 fallback weights where M5 is undefined or zero. We validate retained vote shares, document fallback counties, and compare the results to M6+fallback, pure M2, and M4+fallback. These checks are important because the estimates should not be driven by arbitrary geographic harmonization choices.

The empirical exercise should be interpreted as a reduced-form exposure design, not as definitive evidence of individual belief updating. ADH exposure is a shift-share measure: CZs differ in initial industry composition, and industries differ in later import growth. Identification requires either that initial industry shares are conditionally unrelated to counterfactual political trends, or that the industry-level import shifts are as-good-as-random with respect to local political trends (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2025). These assumptions are not innocuous. In our diagnostics, the ADH instrument strongly predicts exposure, but exposure is also highly correlated with baseline manufacturing intensity and routine-task concentration. Table 3 summarizes the first-stage strength in our political-outcome sample.

Table 3: First-stage diagnostics for ADH exposure.

Specification	Estimate	Std. error	First-stage F	N
Minimal	0.977	0.085	133.4	722
Core controls	0.812	0.101	64.6	722
Full controls	0.794	0.101	62.0	722

Figure 1 presents the main event-study estimates. The preferred inference is CZ-clustered. Conley, Driscoll-Kraay, Newey-West, state-clustered, and two-way clustered alternatives are computed as diagnostics, but several spatial and two-way variance-covariance matrices are non-positive-definite or require numerical repair in this short election-year panel. We therefore state the conclusions cautiously and base the main figure on CZ-clustered confidence intervals.

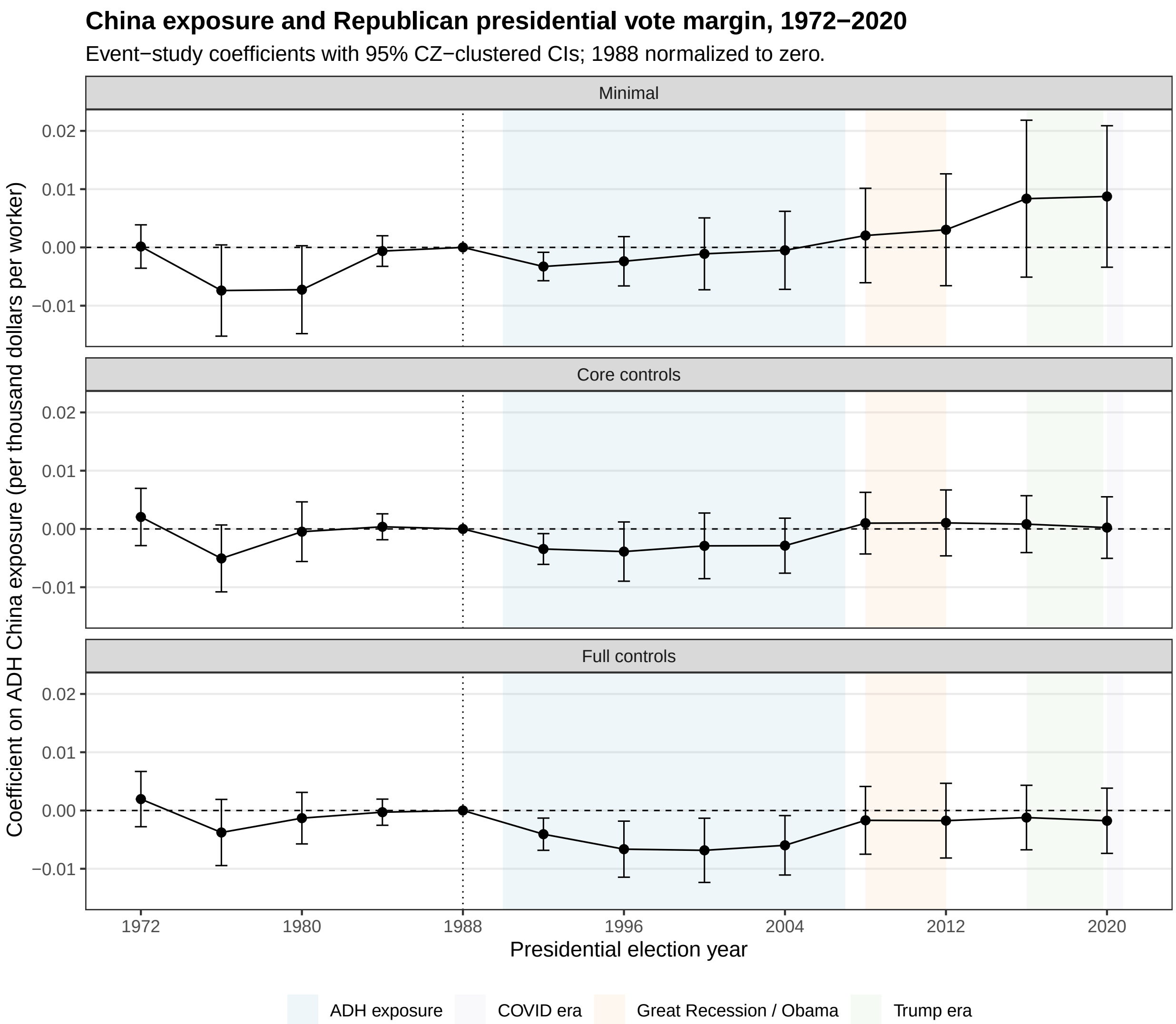

Figure 1: Event-study coefficients for ADH exposure and Republican presidential margin.

The estimates suggest little clear evidence of rising Republican margins before the shock. The post-shock pattern is more consistent with a later political response: the positive association grows in the medium and long run, especially around the 2016 and 2020 horizons. This timing is important because it suggests that the political effects of local trade exposure may not operate immediately through local material losses alone. They may also depend on later national political rhetoric, elite positioning, media narratives, or changing turnout and party coalitions.

To understand what is behind the margin effect, we decompose the outcome into Republican votes per 1990 population, Democratic votes per 1990 population, and two-party vote volume. Figure 2 shows that the margin pattern should not be interpreted mechanically as a simple increase in Republican votes. The broader decomposition helps distinguish vote-choice changes from changes in participation and party-specific mobilization.

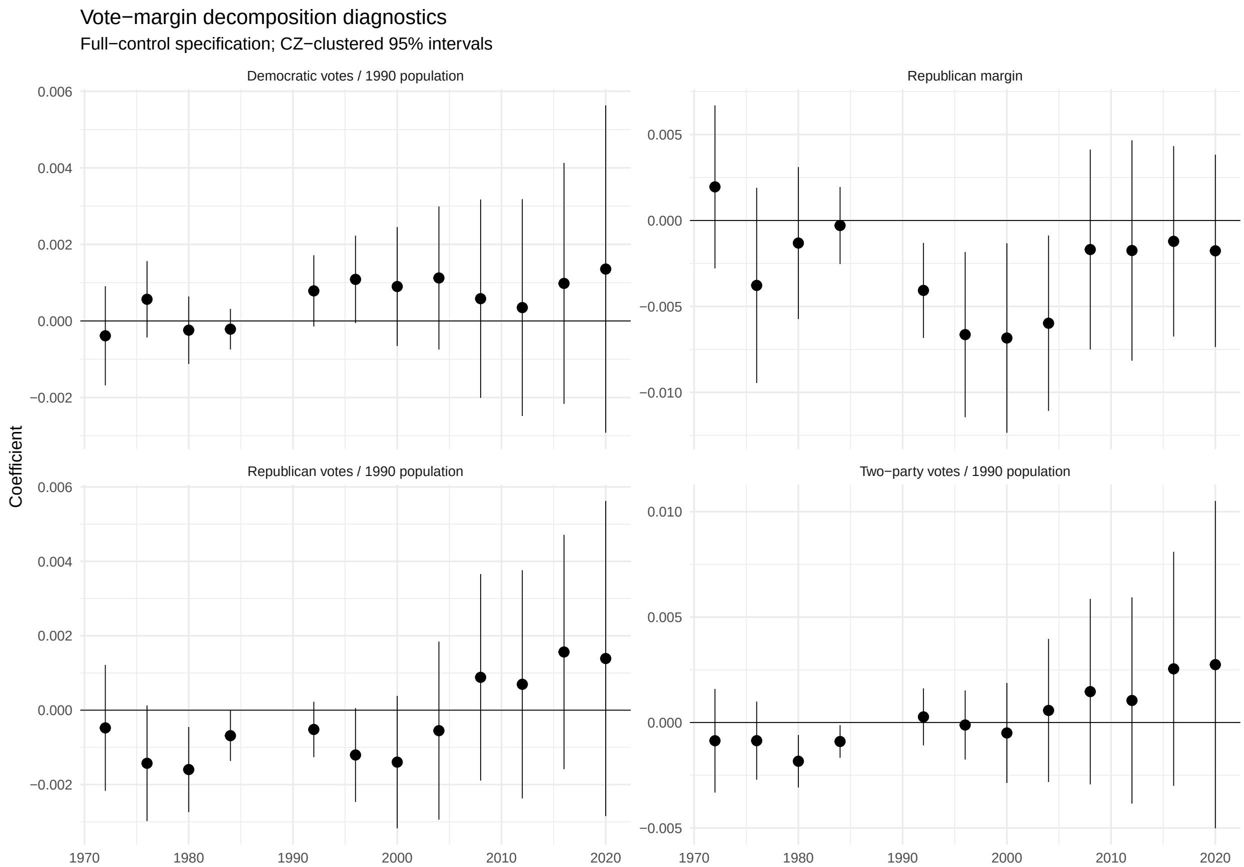

Figure 2: Political-outcome decomposition across election years.

Finally, we examine whether the single 1990–2007 exposure measure masks important timing differences. The original ADH exposure measure aggregates the China shock over a long period, so an event-study coefficient for 2016 does not mean that a new trade shock occurs in 2016. It means that CZs with greater cumulative exposure by 2007 evolved differently by 2016. To make this interpretation more transparent, we split exposure into 1990–2000 and 2000–2007 components and include both components jointly. Figure 3 shows that the estimated relationship varies depending on which subperiod of import growth is emphasized. Pre-1988 coefficients in this exercise are placebo relationships with future exposure, while post-2008 coefficients describe how early- and late-shock exposure predict later political differences.

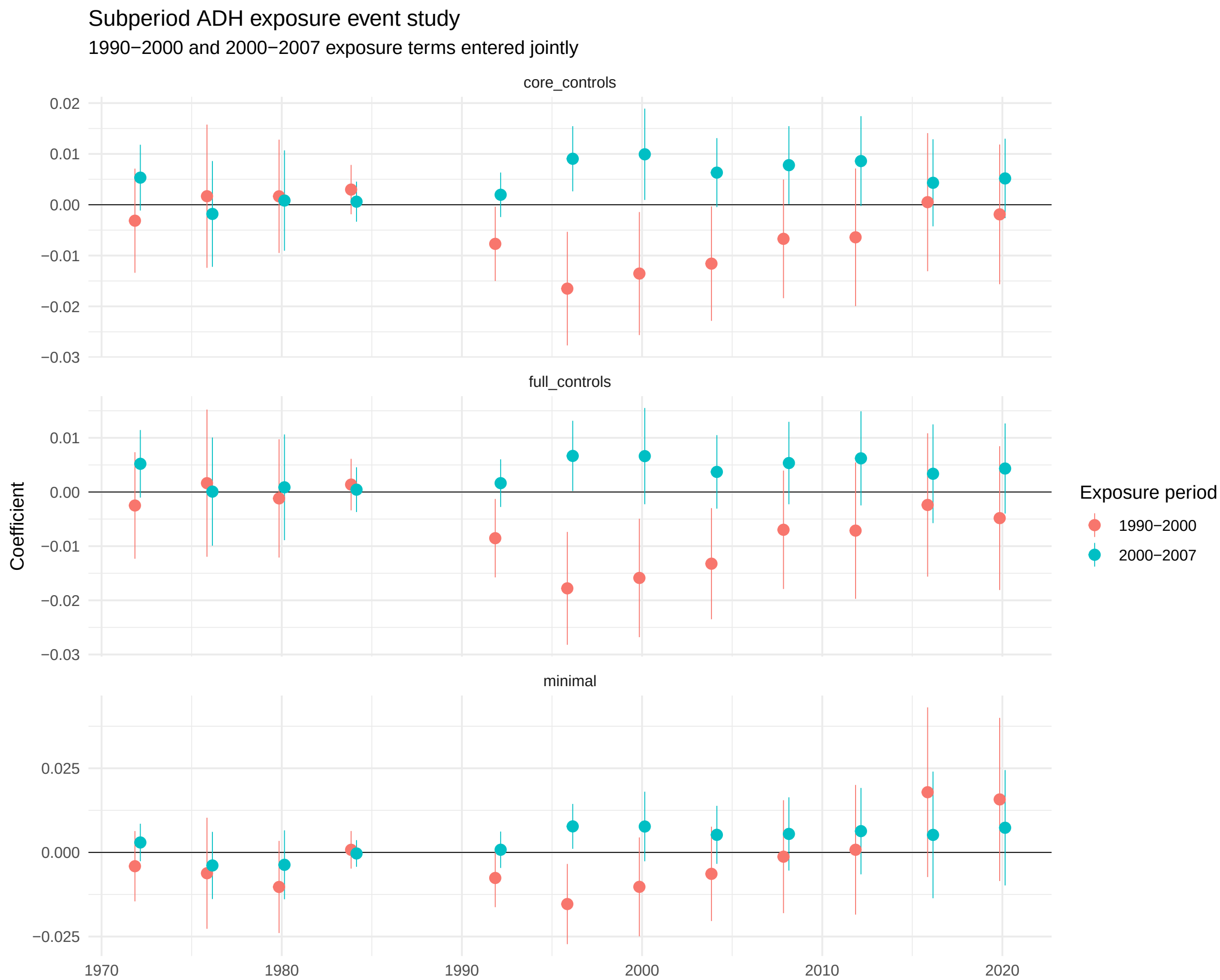

Figure 3: Subperiod ADH exposure event study, with 1990–2000 and 2000–2007 exposure terms entered jointly.

5 Conclusion

Our results suggest that the China shock may have had political consequences beyond its direct effects on manufacturing employment. CZs more exposed to Chinese import competition eventually show more Republican-leaning presidential outcomes, with the clearest differences appearing after the original labor-market shock rather than immediately. This timing makes the interpretation complex. The estimates may reflect voters responding to material local losses, but they may also reflect later elite rhetoric, media framing, turnout changes, candidate supply, or changes in local population composition.

The main lesson is that trade policy cannot be evaluated only through employment and wage effects. Large local labor-market shocks may also affect political representation, polarization, and support for future policy regimes. At the same time, our conclusions should be cautious. The design builds on a Bartik-style exposure measure, so interpretation depends on assumptions about initial industry composition and industry-level shocks. The unit of analysis is a commuting zone, not an individual voter, so we cannot directly observe individual belief updating. Our evidence is most informative about how local political outcomes evolve in places more exposed to import competition, not about a single definitive psychological mechanism.

Future work should therefore focus on mechanism. Turnout and registration data, partisan-index data, migration data, and measures of elite political supply could help distinguish material-experience channels from mobilization, composition, and rhetoric/media channels. Even with these limitations, the evidence suggests that the political effects of globalization are dynamic, geographically concentrated, and potentially mediated by institutions and narratives long after the initial economic shock.

6 References

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A Appendix A: County-to-CZ Harmonization and Measurement

This appendix documents the county-to-CZ bridge used to construct political outcomes. County election returns are bridged to 1990 counties using FTZ crosswalk weights and then aggregated to 1990 CZs. The preferred design uses M5 where valid and M2 fallback where M5 is undefined or zero. The full row-level diagnostics are saved in `output/diagnostics/`; the tables below report compact summaries intended for print.

Table 4: Pipeline validation checks, compact summary.

Check	Status	Fatal	Count
adh one row per czone	pass	TRUE	0
cz lookup unique 1990 counties	pass	TRUE	0
aa all county fips valid	pass	TRUE	0
aa vote identity error within tolerance	pass	TRUE	0
crosswalk positive final weight sums equal one	pass	TRUE	0
crosswalk all source weight sums equal one or reported	reported	FALSE	0
crosswalk undefined selected weights handled	pass	TRUE	1204
aa no duplicate county years after exact distinct	pass	TRUE	0
aa two party votes match party votes	pass	TRUE	0
aa votes nonnegative	pass	TRUE	0
bridged vote retention above threshold	pass	TRUE	0
main panel balanced	pass	TRUE	0
event study residualized design full rank	pass	TRUE	0
event study main vcov available without repair	pass	TRUE	0
event study conley repair diagnostics recorded	pass	FALSE	31

Table 5: Retained vote shares by year and source scope.

Year	Source counties	Fallback counties	Unmatched counties	Retained vote share
1972	3106	258	0	1.0000
1976	3109	258	0	1.0000
1980	3108	221	0	1.0000
1984	3110	221	0	1.0000
1988	3110	221	0	1.0000
1992	3110	0	0	1.0000
1996	3109	0	0	1.0000
2000	3109	196	0	1.0000
2004	3109	196	0	1.0000
2008	3109	196	0	1.0000
2012	3109	183	0	1.0000
2016	3108	183	0	1.0000
2020	3108	184	0	1.0000

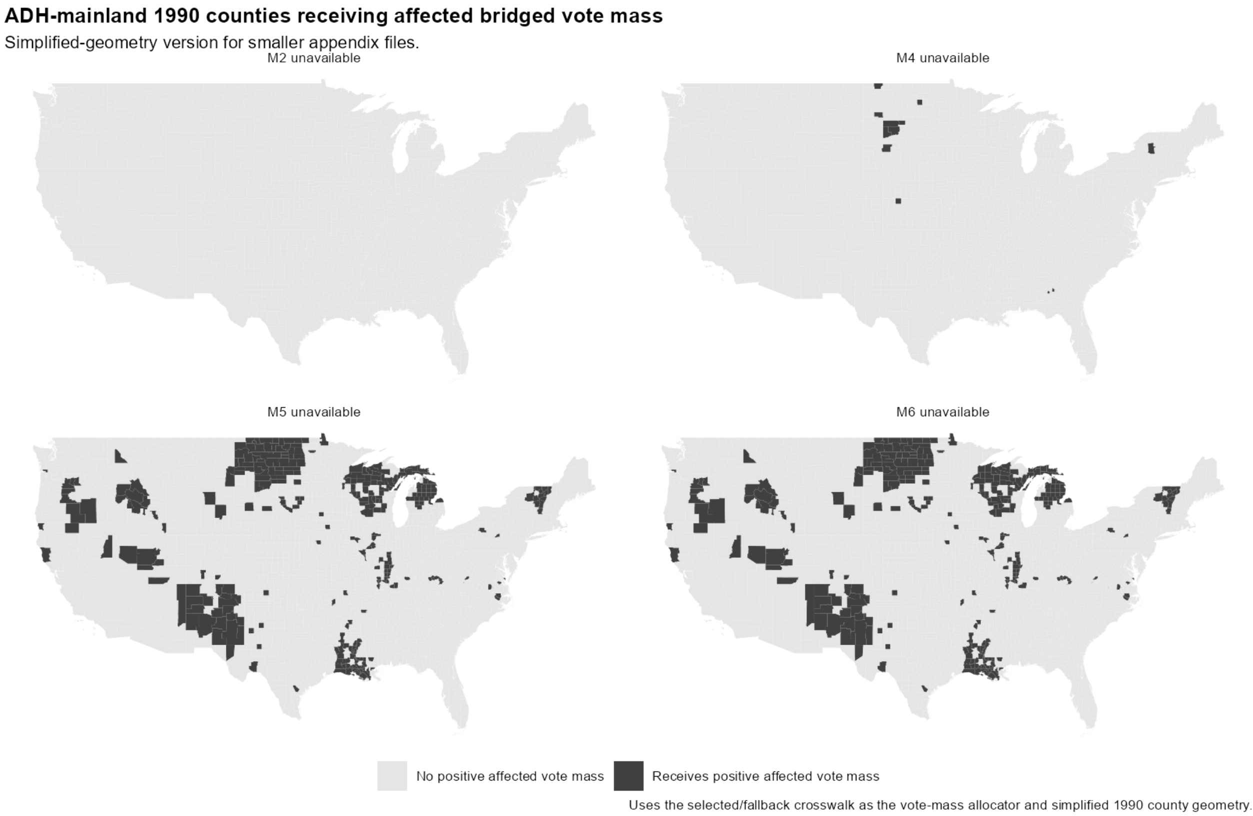

Figure 4: 1990 counties receiving positive vote mass from source counties with unavailable weights.

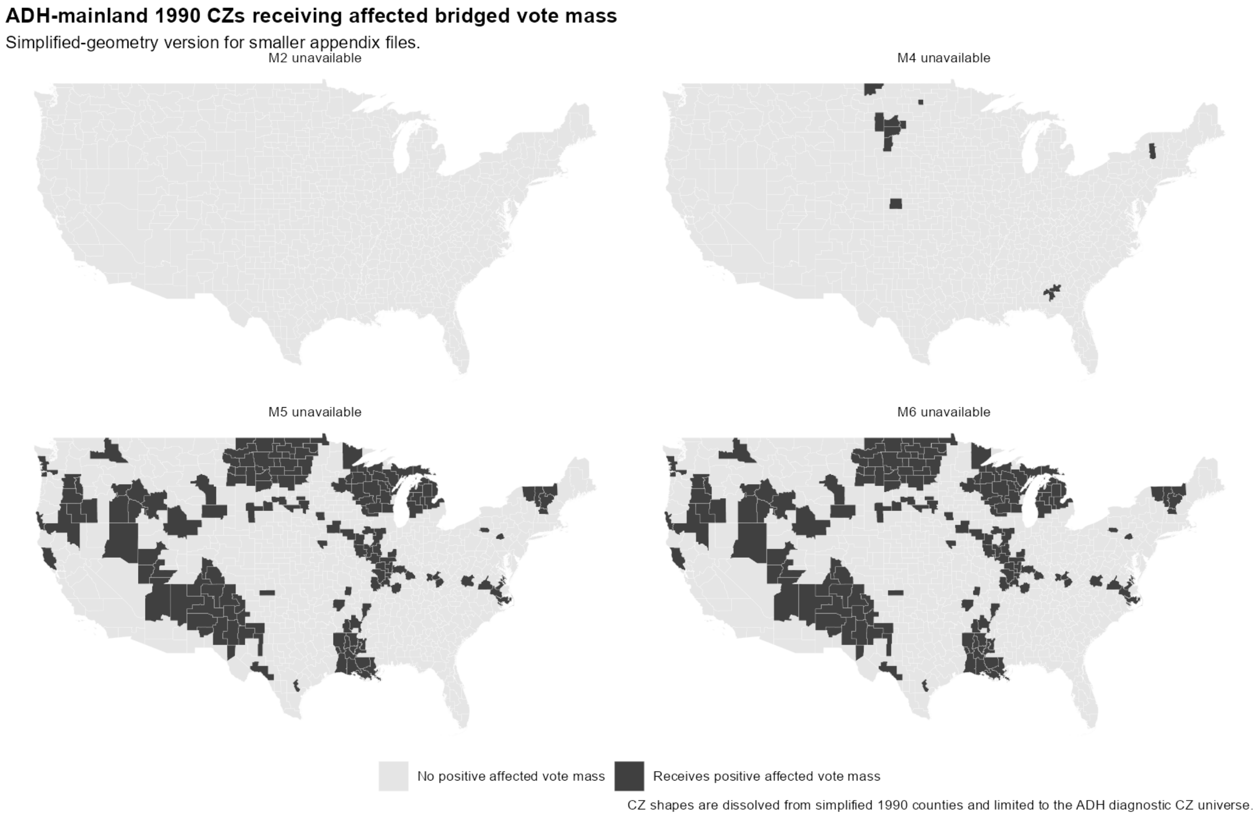

Figure 5: 1990 CZs receiving positive vote mass from source counties with unavailable weights.

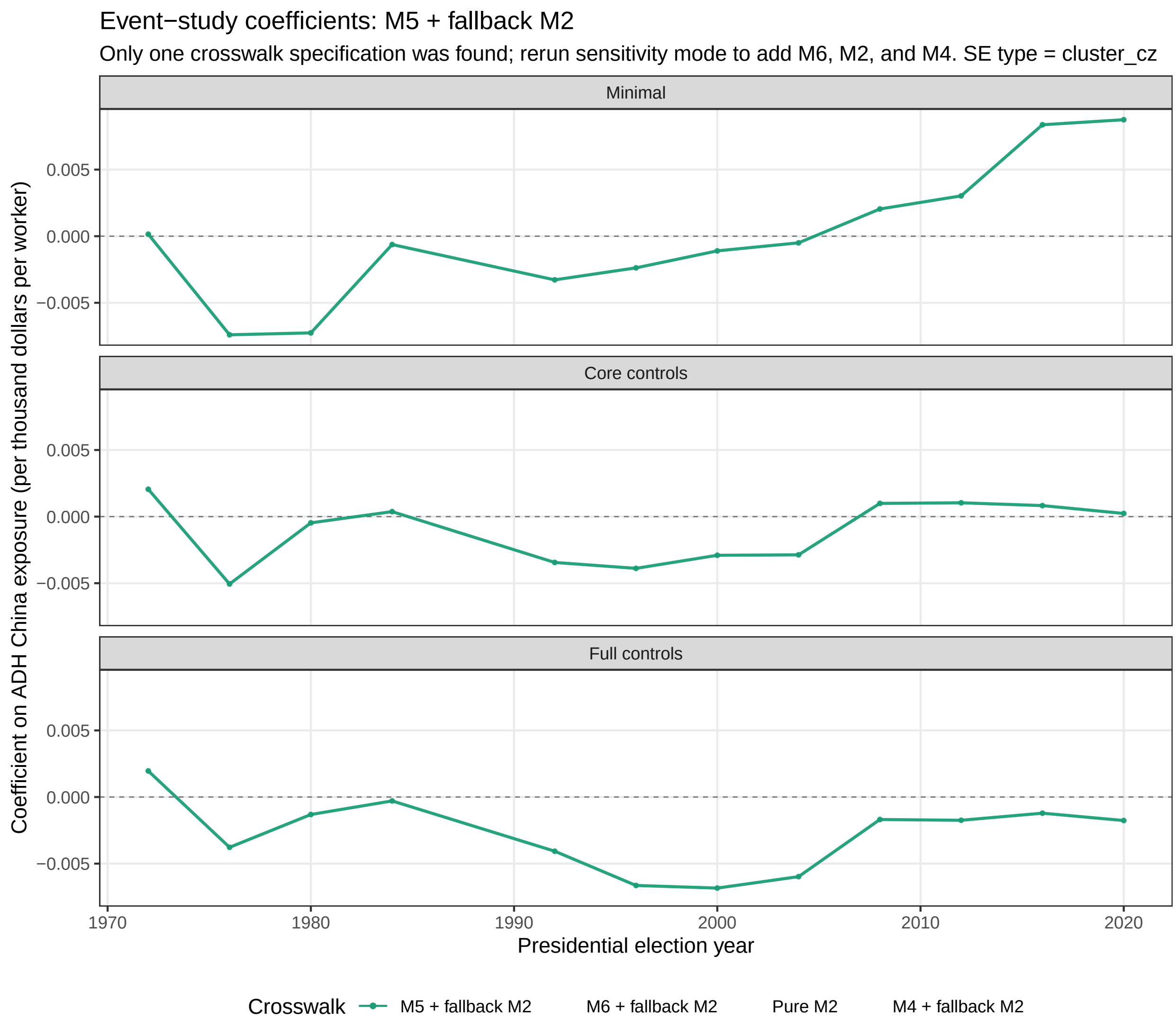

Figure 6: Crosswalk-specification comparison. If the full sensitivity pipeline has not been run, this figure will show only the preferred M5+fallback design.

B Appendix B: Main Event Study and Alternative Political Outcomes

Table 6: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Minimal	1972	0.0002	0.0019	[-0.0036, 0.0039]	0.936
Minimal	1992	-0.0033	0.0012	[-0.0057, -0.0008]	0.009
Minimal	2000	-0.0011	0.0031	[-0.0073, 0.0051]	0.726
Minimal	2008	0.0020	0.0041	[-0.0061, 0.0101]	0.621
Minimal	2016	0.0084	0.0069	[-0.0051, 0.0218]	0.224
Minimal	2020	0.0087	0.0062	[-0.0034, 0.0209]	0.159

Table 6: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Core controls	1972	0.0021	0.0025	[-0.0029, 0.0070]	0.412
Core controls	1992	-0.0034	0.0013	[-0.0061, -0.0008]	0.011
Core controls	2000	-0.0029	0.0029	[-0.0085, 0.0027]	0.313
Core controls	2008	0.0010	0.0027	[-0.0043, 0.0063]	0.713
Core controls	2016	0.0008	0.0025	[-0.0041, 0.0057]	0.740
Core controls	2020	0.0002	0.0027	[-0.0050, 0.0055]	0.930
Full controls	1972	0.0020	0.0024	[-0.0028, 0.0067]	0.419
Full controls	1992	-0.0041	0.0014	[-0.0068, -0.0013]	0.004
Full controls	2000	-0.0068	0.0028	[-0.0123, -0.0013]	0.015
Full controls	2008	-0.0017	0.0030	[-0.0075, 0.0041]	0.569
Full controls	2016	-0.0012	0.0028	[-0.0068, 0.0043]	0.668
Full controls	2020	-0.0018	0.0029	[-0.0074, 0.0038]	0.537

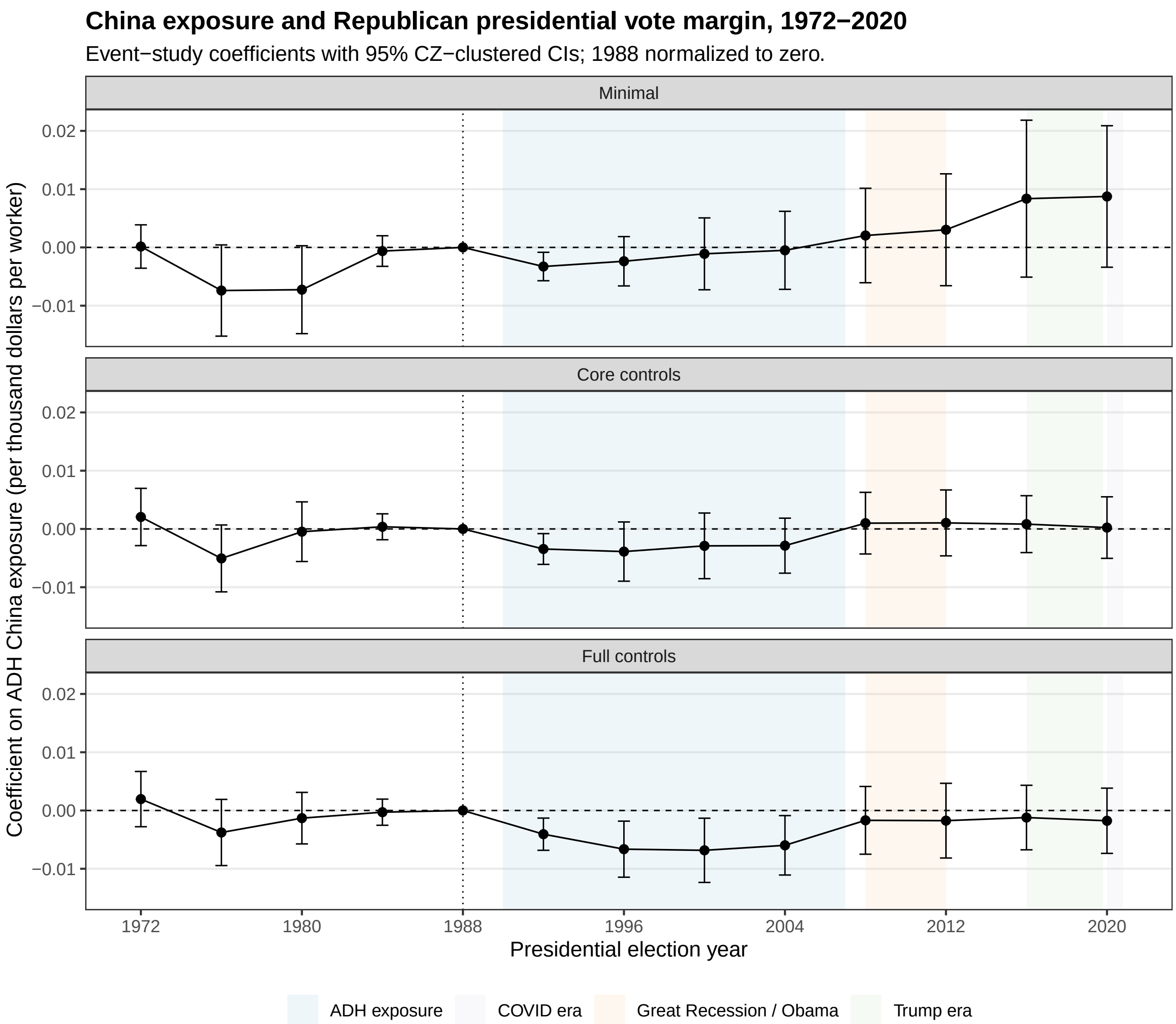

Figure 7: Main event-study estimates.

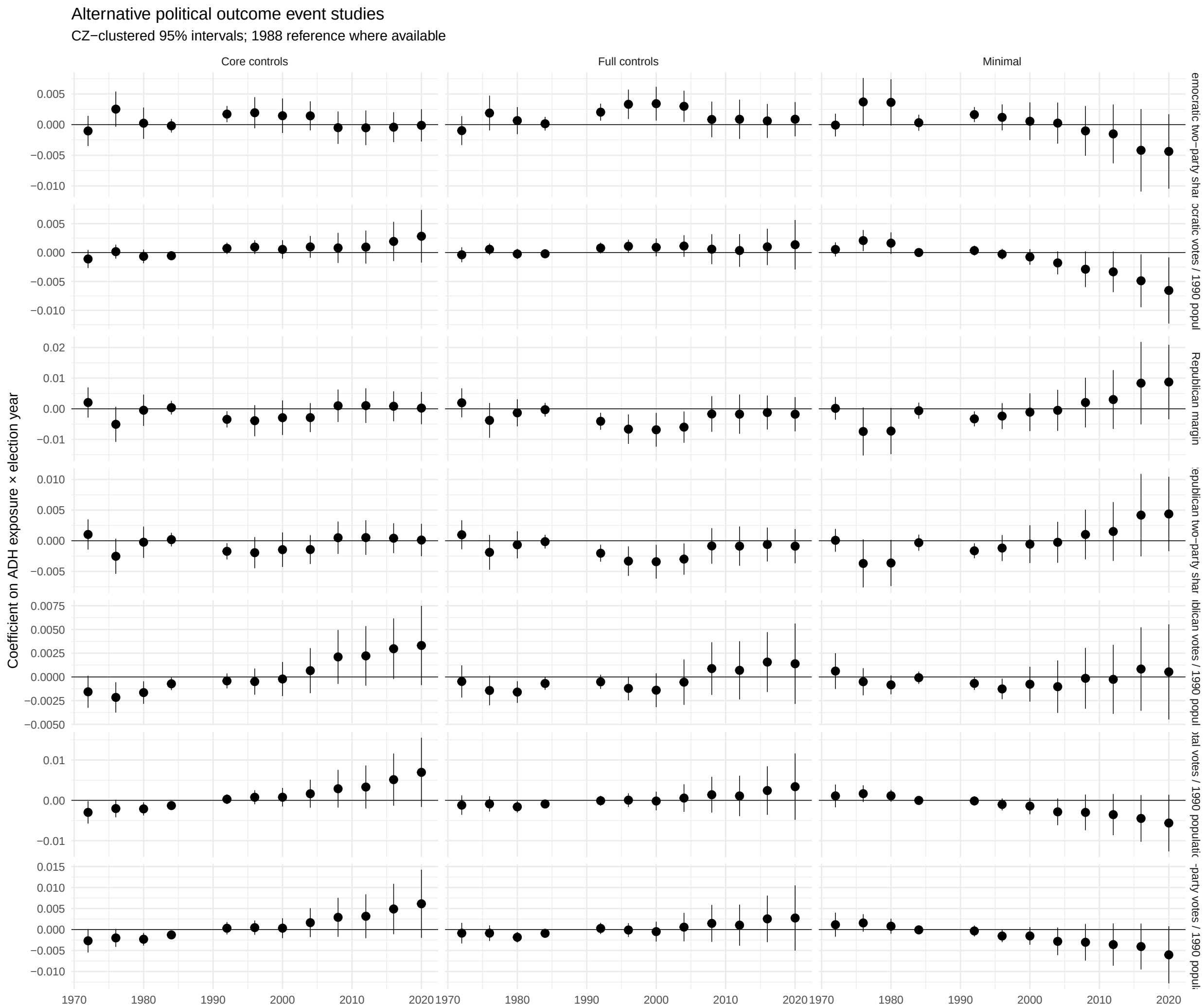

Figure 8: Full alternative-outcome event-study grid.

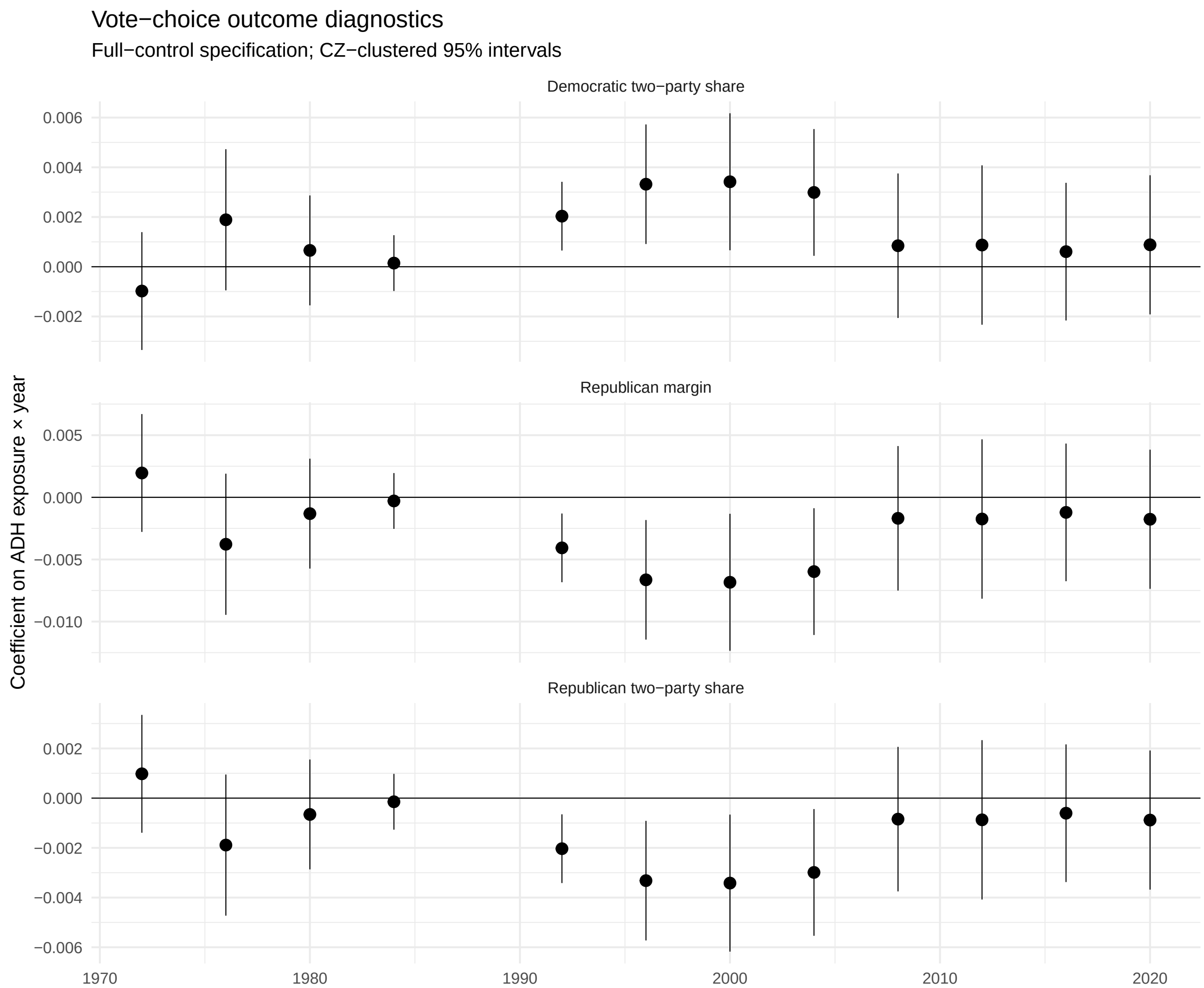

Figure 9: Vote-choice outcome family.

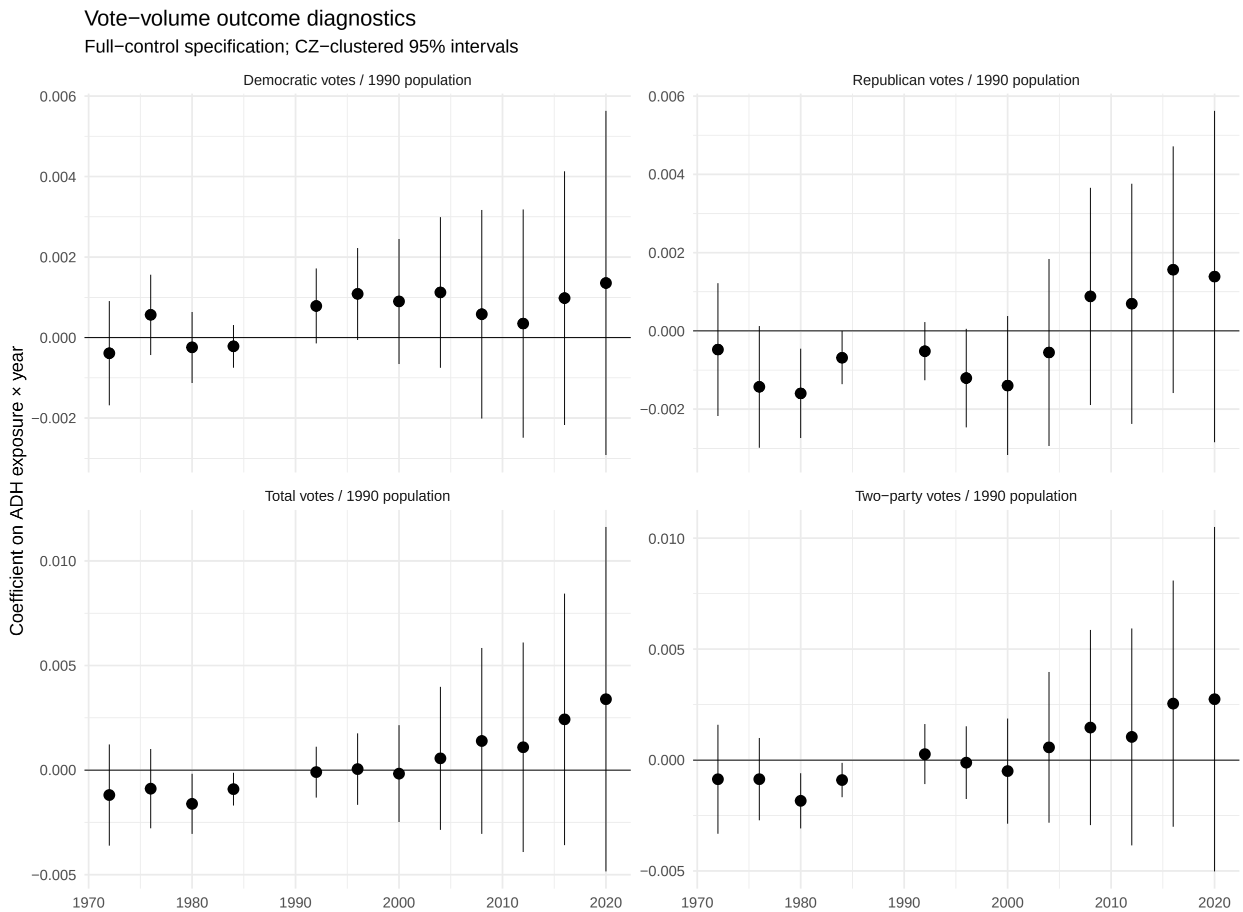

Figure 10: Vote-volume outcome family.

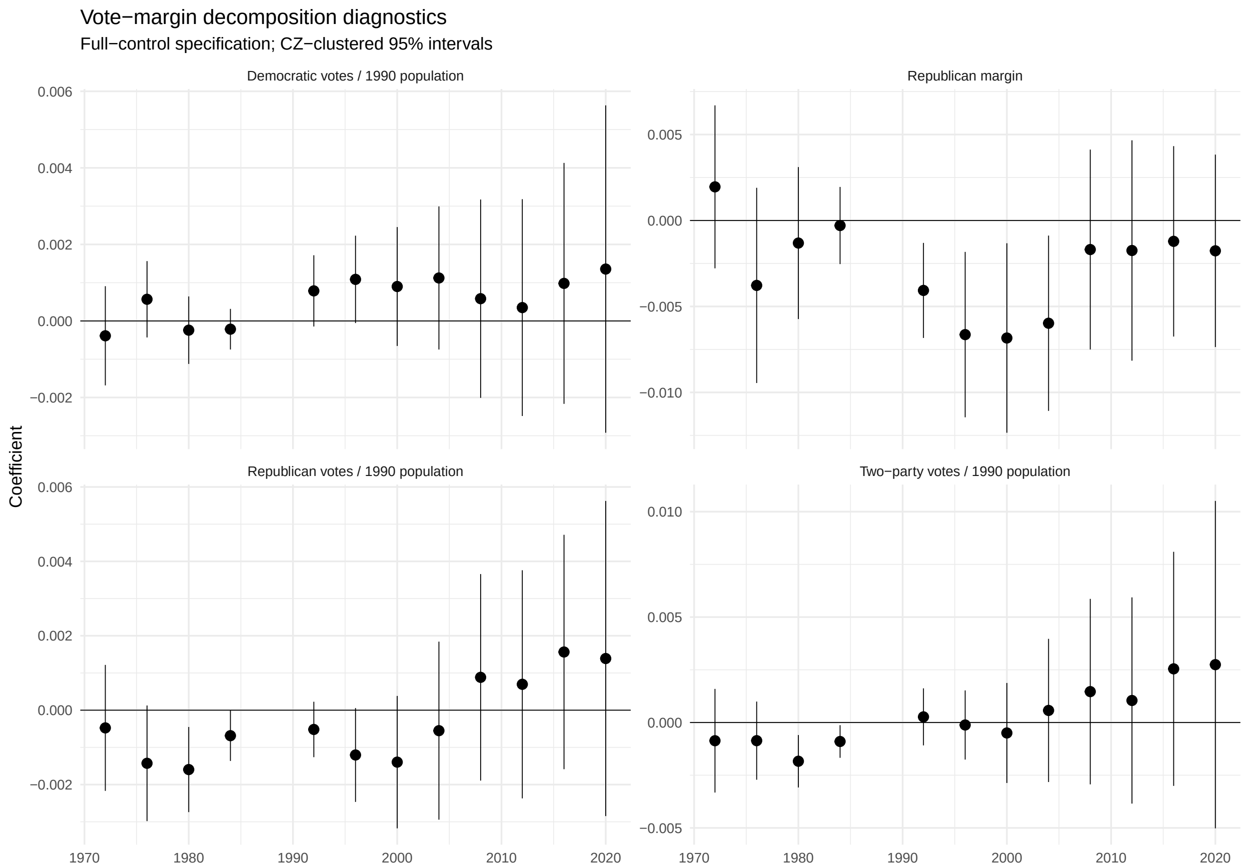

Figure 11: Outcome decomposition.

C Appendix C: Bartik Identification Diagnostics

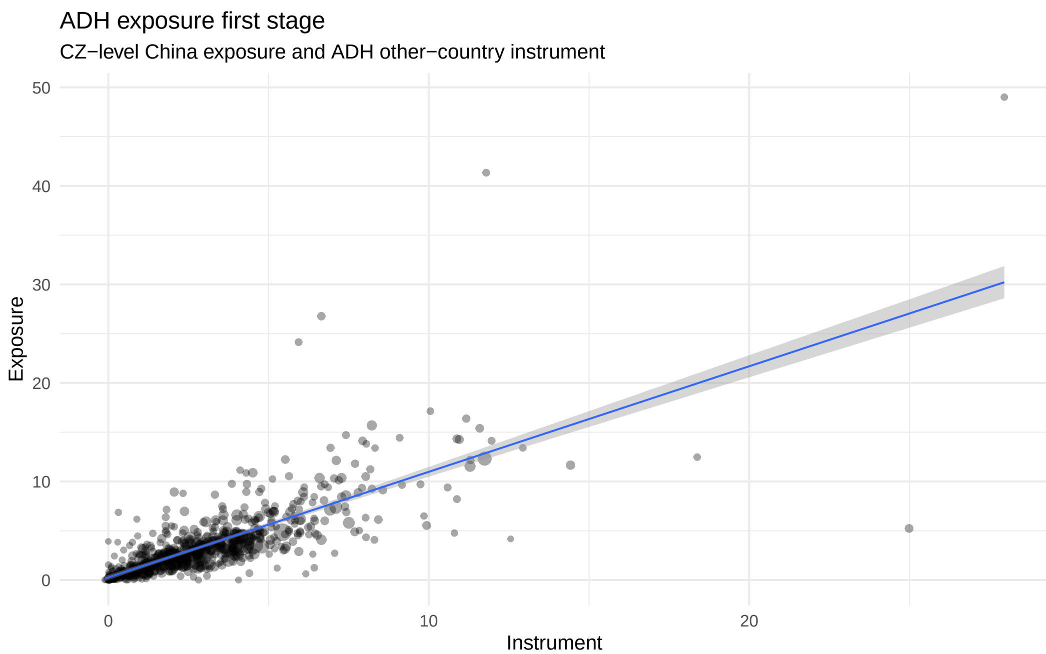

Figure 12: First-stage relationship between ADH exposure and the ADH instrument.

Table 7: Baseline balance: correlations with exposure and instrument.

Variable	Corr. with exposure	Corr. with instrument	Weighted mean	SD
l shind manuf cbp	0.574	0.614	20.695	12.848
l sh popedu c	-0.243	-0.287	48.021	8.574
l sh popfborn	-0.145	-0.131	10.146	4.969
l sh empl f	0.028	0.006	63.624	6.634
l sh routine33	0.301	0.355	32.228	3.362
l task outsource	0.261	0.312	0.112	0.407
statefip	-0.085	-0.087	27.790	14.987

Table 8: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend slope rep margin	Minimal	exposure 1990 2007	0.0001	0.0002	0.601
pretrend slope rep margin	Core controls	exposure 1990 2007	-0.0001	0.0002	0.700
pretrend slope rep margin	Full controls	exposure 1990 2007	-0.0001	0.0002	0.548
pretrend slope rep margin	Minimal	instrument 1990 2007	0.0000	0.0002	0.976

Table 8: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend slope rep margin	Core controls	instrument 1990 2007	-0.0001	0.0002	0.556
pretrend slope rep margin	Full controls	instrument 1990 2007	-0.0002	0.0002	0.498
pretrend change first to last	Minimal	exposure 1990 2007	0.0075	0.0062	0.236
pretrend change first to last	Core controls	exposure 1990 2007	-0.0012	0.0056	0.833
pretrend change first to last	Full controls	exposure 1990 2007	-0.0018	0.0061	0.769
pretrend change first to last	Minimal	instrument 1990 2007	0.0064	0.0083	0.445
pretrend change first to last	Core controls	instrument 1990 2007	-0.0001	0.0081	0.990
pretrend change first to last	Full controls	instrument 1990 2007	-0.0004	0.0087	0.959

Table 9: Industry-shift summary from ADH trade-data diagnostics.

Status	Period	Importer	SIC87DD	Start imports	End imports	Import growth
ok	1991-1999	OTH	3679	143699.7	2505878.0	2362178.3
ok	1991-1999	OTH	3577	40964.0	1565398.4	1524434.3
ok	1991-1999	OTH	3944	1285590.8	2732429.2	1446838.5
ok	1991-1999	OTH	3651	936563.7	2377214.8	1440651.1
ok	1991-1999	OTH	3861	58608.0	1004058.8	945450.8
ok	1991-1999	OTH	2369	1191708.4	2126562.8	934854.4
ok	1991-1999	OTH	2337	1214870.9	2095917.0	881046.1
ok	1991-1999	OTH	2091	235925.2	1114860.0	878934.8
ok	1991-1999	OTH	3161	623382.1	1466893.6	843511.5
ok	1991-1999	OTH	3089	256498.7	1082972.6	826474.0
ok	1991-1999	OTH	2325	634508.3	1300735.5	666227.2
ok	1991-1999	OTH	2392	605945.9	1227629.4	621683.4

D Appendix D: Subperiod Exposure and Dynamic Interpretation

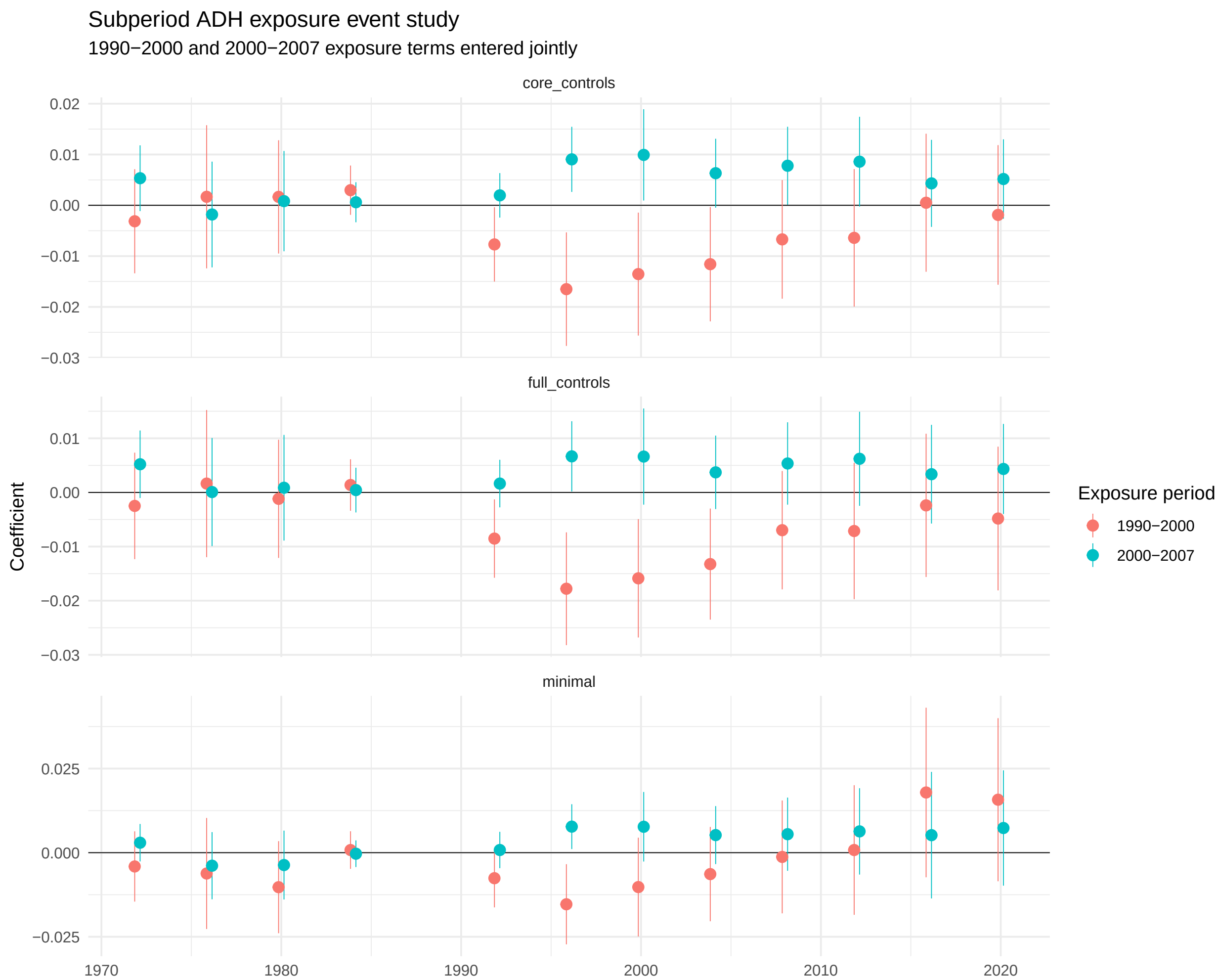

Figure 13: Subperiod exposure event-study estimates.

Table 10: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late	Approx. p-value
Minimal	1996	-0.0153	0.0077	-0.0231	0.001
Minimal	2000	-0.0102	0.0077	-0.0179	0.050
Minimal	2008	-0.0013	0.0055	-0.0068	0.508
Minimal	2016	0.0179	0.0052	0.0127	0.429
Minimal	2020	0.0157	0.0073	0.0084	0.579
Core controls	1996	-0.0165	0.0090	-0.0256	0.000
Core controls	2000	-0.0135	0.0099	-0.0235	0.002

Table 10: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late	Approx. p-value
Core controls	2008	-0.0067	0.0078	-0.0145	0.042
Core controls	2016	0.0005	0.0043	-0.0038	0.642
Core controls	2020	-0.0019	0.0052	-0.0071	0.381
Full controls	1996	-0.0178	0.0067	-0.0245	0.000
Full controls	2000	-0.0159	0.0066	-0.0225	0.002
Full controls	2008	-0.0070	0.0053	-0.0123	0.070
Full controls	2016	-0.0024	0.0034	-0.0058	0.482
Full controls	2020	-0.0048	0.0043	-0.0092	0.252

E Appendix E: Standard-Error Diagnostics

The main specification uses CZ-clustered standard errors. The diagnostics below summarize the broader set of variance-covariance estimators computed by the pipeline. Full row-level diagnostics are available in `output/diagnostics/event_study_vcov_diagnostics.csv` and `output/diagnostics/event_study_se_warning_diagnostics.csv`.

Table 11: Variance-covariance diagnostics by SE type.

SE type	status	Model rows	Repairs needed	Max negative eigenvalues	Smallest eigenvalue
CZ x year	repair_required	8	8	94	-0.004037
CZ-clustered	ok	8	0	0	0.000000
Conley 1000 km	repair_required	8	8	57	-0.005516
Conley 250 km	ok	1	0	0	0.000000
Conley 250 km	repair_required	7	7	39	-0.000221
Conley 500 km	repair_required	8	8	54	-0.001216
Conley 750 km	repair_required	8	8	56	-0.002515
Driscoll-Kraay	ok	8	0	51	-0.000000
HC	ok	8	0	0	0.000000
Newey-West	ok	8	0	0	0.000000
State-clustered	ok	8	0	37	-0.000000

Table 12: SE diagnostic issue summary.

SE type	Issue	Model rows	Max near-zero SEs	Max nonfinite SEs	Max rank deficiency
CZ x year	Two-way cluster: few year clusters	8	0	10	0
Conley 1000 km	Conley VCOV non-PD or repaired	8	0	1	0
Conley 250 km	Conley VCOV non-PD or repaired	7	0	0	0
Conley 500 km	Conley VCOV non-PD or repaired	8	0	0	0
Conley 750 km	Conley VCOV non-PD or repaired	8	0	1	0
Driscoll-Kraay	Near-zero or nonfinite SEs	8	17	0	69
State-clustered	Rank-deficient VCOV	5	0	0	72

F Appendix F: Exploratory Mechanism Evidence

F.1 NaNDA turnout, vote ratios, and partisanship

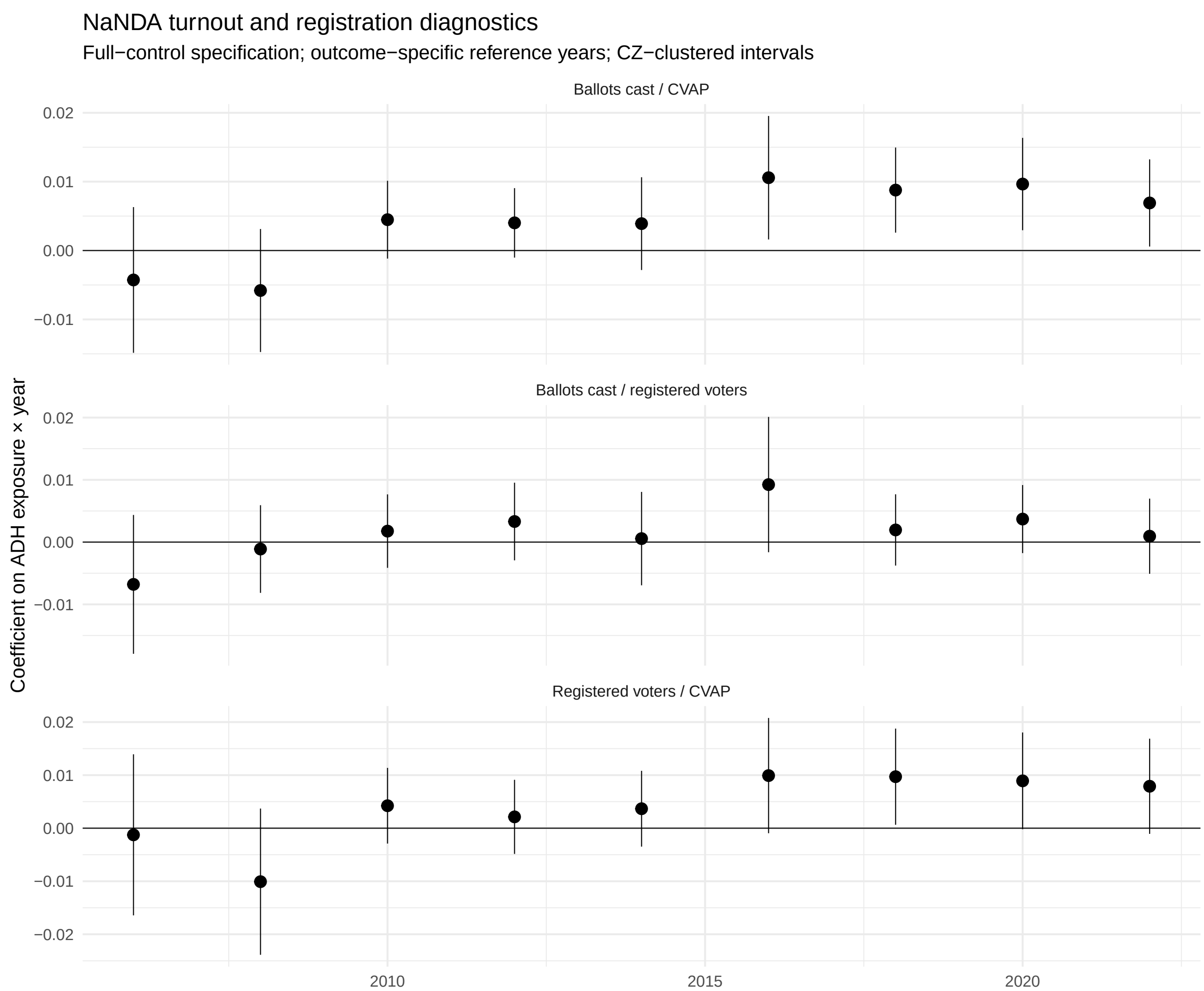

Figure 14: NaNDA turnout and registration outcomes.

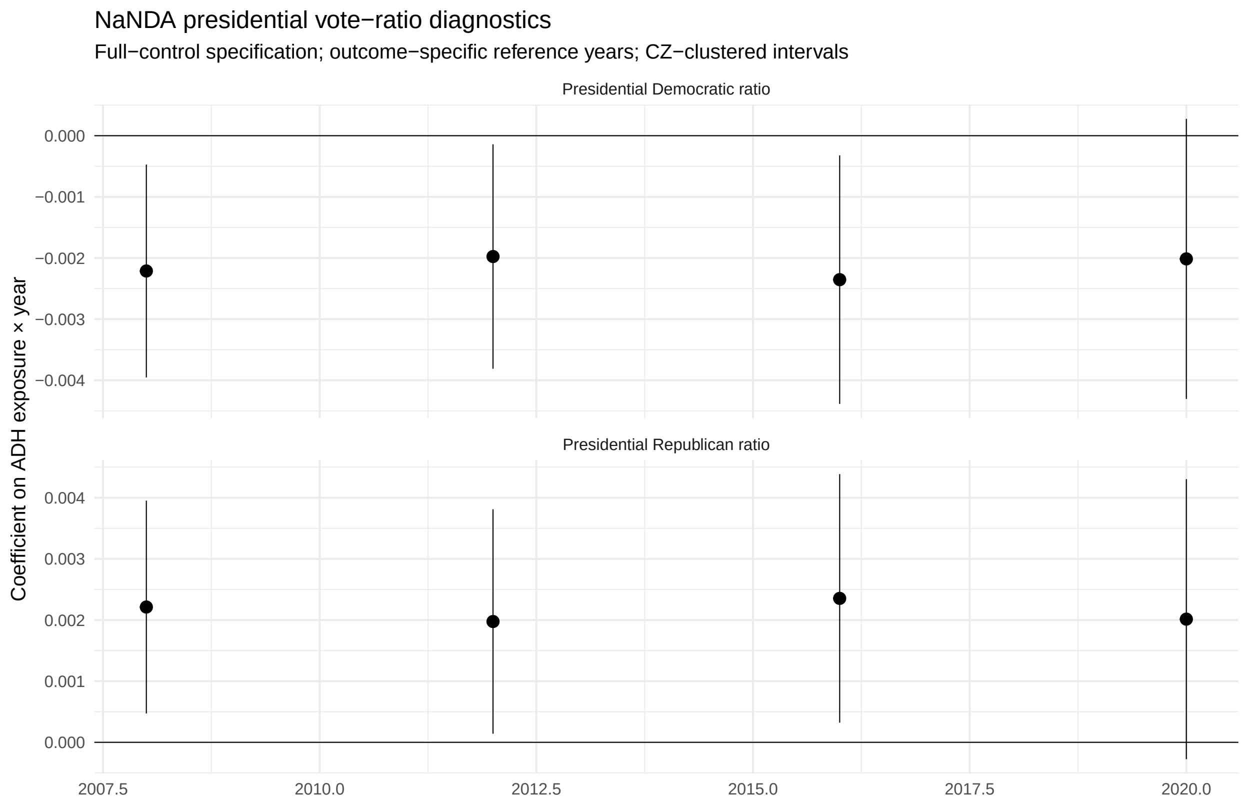

Figure 15: NaNDA presidential vote ratios.

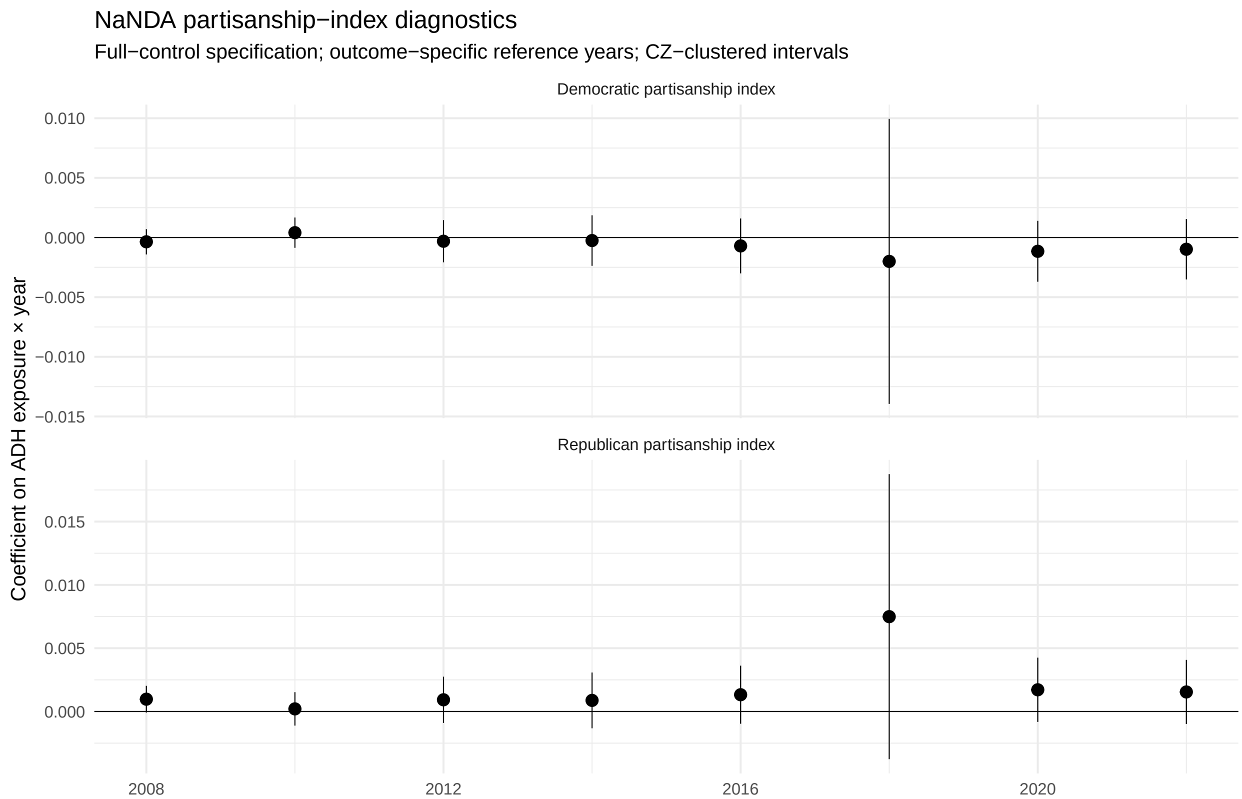

Figure 16: NaNDA partisanship indices.

Table 13: NaNDA outcome availability summary.

Outcome	Valid years	Unavailable years	Min non-missing CZs	Years with all-zero values
Ballots / CVAP	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	721	None
Ballots / registered voters	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	661	None
Democratic partisanship index	2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	2004	722	2004
Presidential Democratic ratio	2004, 2008, 2012, 2016, 2020	2006, 2010, 2014, 2018, 2022	722	None
Presidential Republican ratio	2004, 2008, 2012, 2016, 2020	2006, 2010, 2014, 2018, 2022	722	None
Registered / CVAP	2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	None	720	None

Table 13: NaNDA outcome availability summary.

Outcome	Valid years	Unavailable years	Min non-missing CZs	Years with all-zero values
Republican partisanship index	2006, 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022	2004	722	2004

F.2 ADHM 2020 public political-supply diagnostics

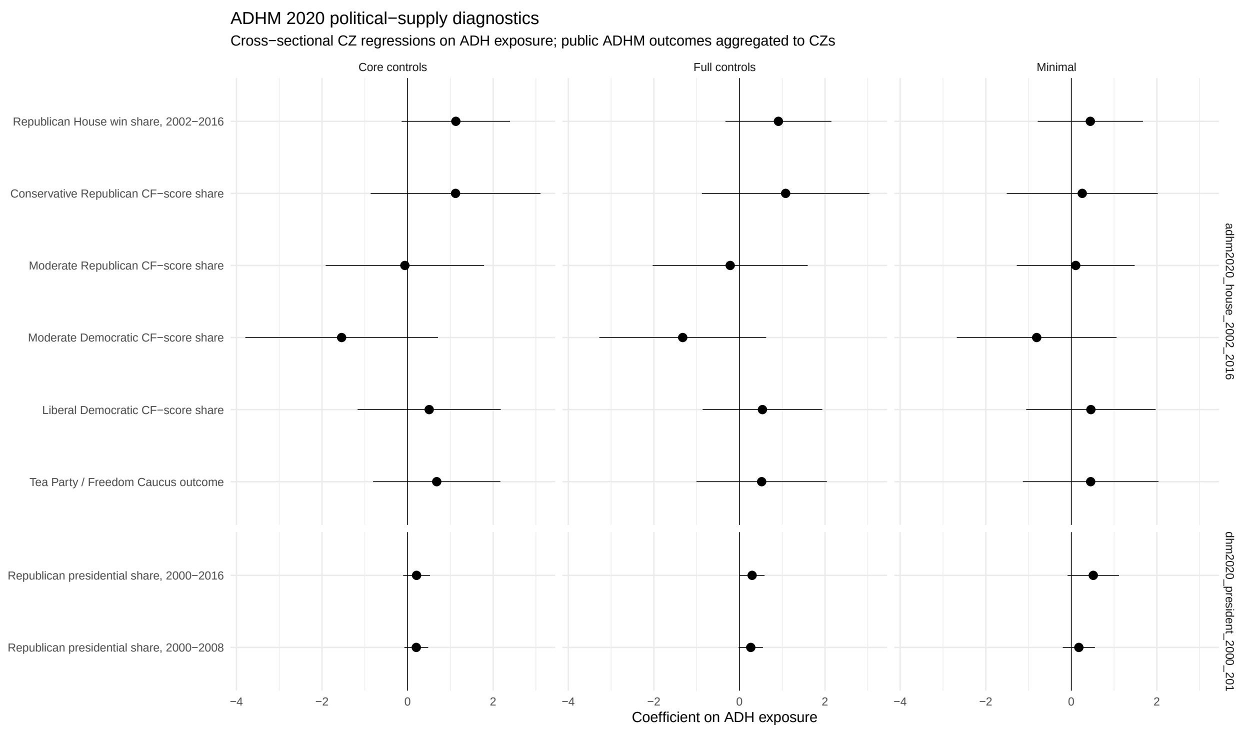

Figure 17: ADHM 2020 public political-supply mechanism regressions.

Table 14: ADHM 2020 mechanism regressions, full-control specification.

Outcome	Estimate	Std. error	p-value	N
Republican House win share, 2002-2016	0.911	0.632	0.156	722
Conservative Republican CF-score share	1.080	0.999	0.285	722
Moderate Republican CF-score share	-0.216	0.925	0.816	722
Moderate Democratic CF-score share	-1.326	0.995	0.189	722
Liberal Democratic CF-score share	0.538	0.714	0.455	722
Tea Party / Freedom Caucus outcome	0.520	0.778	0.508	722
Republican presidential share, 2000-2016	0.296	0.148	0.052	722
Republican presidential share, 2000-2008	0.265	0.145	0.075	722

G Appendix G: Replication Notes

The complete analysis can be regenerated from the project root with:

```r
source(here::here("src", "run_pipeline.R"))
```

For the final appendix-style crosswalk-sensitivity run:

```r
Sys.setenv(RUN_CROSSWALK_SENSITIVITY = "true")
source(here::here("src", "run_pipeline.R"))
```

The replication-code PDF can be generated from the `report` folder with:

```r
source("make_replication_code_pdf.R")
```